SimpleTM++: Adaptive Hybrid Tokenization for Multivariate Time Series Forecasting

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by

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CANDIDATE DECLARATION

I hereby certify that the work, which is being presented in the report, entitled **Connected Component Analysis for Multi-Vehicle Detection in Shipment Tracking Systems**, in fulfillment of the requirement for the award of the Degree of **Bachelor of Technology (B.Tech)** and submitted to the institution is an authentic record of my work carried out during the period *January 2026* to *March 2026* under the supervision of **Dr. Anshul**. I have also cited the reference to the text(s)/figure(s)/table(s) from where they have been taken.

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This is to certify that the above statement made by the candidate is true to the best of my knowledge.

Date: _____ Signature of the Project Supervisor

Name: **Dr. Anshul**

Abstract

Multivariate Time Series Forecasting (MTSF) is a fundamental problem with wide applications in energy systems, climate modelling, financial analysis, and traffic monitoring. The SimpleTM framework [1] proposes a lightweight yet competitive architecture that employs Stationary Wavelet Transform (SWT) tokenization and a Clifford-algebra-inspired geometric attention mechanism. However, its fixed single-strategy tokenisation may not be optimal across datasets exhibiting heterogeneous signal characteristics.

This project—**SimpleTM++**—systematically investigates whether replacing or extending the original SWT tokeniser with an FFT-based tokeniser, or combining multiple tokenisation strategies through adaptive fusion, can improve forecasting robustness and accuracy. Work completed at mid-semester includes: a thorough literature review, full understanding of the SimpleTM codebase, multi-dataset exploratory data analysis (EDA), implementation of both the `SimpleTM_SWT` baseline and the `SimpleTM_FFT` variant, and experimental evaluation on two ETT benchmark datasets. On ETTm1 (seq = 96, pred = 96), `SimpleTM_SWT` achieves MSE = 0.3497 while `SimpleTM_FFT` achieves MSE = 0.3630—an unexpected gap given the strong frequency-domain structure observed in EDA. The report analyses this discrepancy in depth and motivates why a hybrid tokenisation approach remains the correct long-term direction. The planned `SimpleTM++` hybrid model fuses SWT, FFT, and local convolutional branches through learnable gating, and is expected to overcome the limitations of each individual tokeniser.

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Chapter 1

Introduction

1.1 Background

Time-series forecasting is the task of predicting future values of a signal using historical observations. In the *multivariate* setting, N variables evolve jointly over time and the forecaster must exploit both temporal dependencies within each channel and inter-channel correlations. Accurate multivariate forecasting underpins a broad set of high-impact applications:

- Electricity demand and load estimation [5];
- Weather and climate prediction;
- Traffic flow and road-occupancy monitoring;
- Industrial sensor analysis and anomaly detection;
- Financial price and volatility modelling.

Deep learning, particularly Transformer-based models, has achieved state-of-the-art performance on standard MTSF benchmarks. However, large Transformer models—including LLM-adapter variants and architectures such as iTransformer and PatchTST—require substantial computational resources and extensive domain-specific fine-tuning. This limits their applicability in resource-constrained or real-time settings.

1.2 Motivation

SimpleTM [1] was proposed as a *lightweight, specialised* alternative. By combining SWT-based tokenisation with geometric attention (combining dot-product and exterior product scores), even a one- or two-layer model achieves results competitive with much larger architectures on standard benchmarks. This raises a deeper question: is the SWT tokenisation the *optimal* choice for all datasets, or can other signal-processing representations provide complementary or superior inductive biases?

EDA on the ETT datasets reveals that real-world time series exhibit qualitatively different structures:

- **Strong periodic components** visible in FFT spectra (motivating FFT tokenisation);
- **Multi-scale non-stationary dynamics** captured by wavelet decomposition (motivating SWT tokenisation);
- **Short-range local motifs** and abrupt spikes (motivating convolutional tokenisation).

No single tokeniser is universally optimal; this motivates the **SimpleTM++** hybrid framework proposed in this project.

1.3 Scope and Contributions of This Project

This B.Tech Project (BTP) is an extension of the official SimpleTM repository (https://github.com/vsingh-group/SimpleTM). The project repository is available at https://github.com/agamswami/SimpleTMG. The specific contributions are:

1. Systematic multi-dataset EDA justifying the need for multiple tokenisation strategies.
2. Clean re-implementation of the original SimpleTM as `SimpleTM_SWT` for fair ablation.
3. Novel `SimpleTM_FFT` variant: replaces SWT with FFT spectral-band tokenisation while keeping geometric attention identical.
4. Controlled experimental comparison on ETTh1 and ETTm1 benchmark datasets.
5. Analysis of the surprising FFT underperformance on ETTm1 despite favourable EDA signals.
6. Design of the planned `SimpleTM++` hybrid architecture.

1.4 Report Structure

Chapter 2 reviews related work. Chapter 3 states the problem formally. Chapter 4 describes the methodology. Chapter 5 details the system architecture. Chapter 6 presents EDA findings. Chapter 7 covers experimental setup and results. Chapter 8 discusses key findings, the FFT–SWT gap, and the hybrid rationale. Chapter 9 enumerates current limitations. Chapter 10 outlines planned work. Chapter 11 concludes.

Chapter 2

Literature Review

2.1 Multivariate Time Series Forecasting

Multivariate time series forecasting has been addressed through a wide range of modelling paradigms. Classical statistical methods such as ARIMA and VAR provide interpretable models but fail to capture complex nonlinear dependencies. Recurrent neural networks (RNNs) and their gated variants (LSTM, GRU) model sequential dependencies but suffer from gradient issues over long horizons.

Transformer-based models [5] have transformed sequence modelling broadly. In the time-series domain, Informer [6] introduced sparse attention for long-sequence forecasting. Autoformer [7] incorporated a decomposition block (trend + seasonality) into the Transformer structure. FEDformer leveraged frequency-enhanced attention. PatchTST [8] introduced patch-based tokenisation to better capture local temporal semantics. TimeMixer combined multi-scale mixing of seasonal and trend information. iTransformer inverted the attention axis to operate over variate tokens rather than time steps.

A common challenge across all these models is that they are relatively large and computationally expensive, requiring significant memory and often dataset-specific fine-tuning.

2.2 SimpleTM

SimpleTM [1] was accepted at ICLR 2025 and proposes a radically lightweight alternative. Its key innovations are:

1. **Inverted embedding**: treats each variate as a token, mapping $\mathbf{x} \in \mathbb{R}^{B \times L \times N}$ to $\mathbf{z} \in \mathbb{R}^{B \times N \times d}$ via a linear projection over time.
2. **SWT tokenisation**: applies the Stationary Wavelet Transform (SWT) to decompose each token into $m + 1$ frequency-scale sub-bands, producing multi-scale representations without downsampling.
3. **Geometric attention**: replaces standard dot-product attention with a Clifford-algebra-inspired score combining a dot product and a wedge product (exterior product magnitude):

$$\text{score}(\mathbf{q}, \mathbf{k}) = (1 - \alpha)\, \mathbf{q}^\top \mathbf{k} + \alpha\, \|\mathbf{q} \wedge \mathbf{k}\|,$$

   where $\|\mathbf{q} \wedge \mathbf{k}\| = \sqrt{\|\mathbf{q}\|^2 \|\mathbf{k}\|^2 - (\mathbf{q}^\top \mathbf{k})^2}$, giving high scores to both similar and geometrically diverse token pairs.
4. **Learnable ISWT**: a trainable inverse SWT reconstructs the output signal from multi-scale attention outputs.

The OpenReview peer assessments [2] highlight several strengths: simplicity, low memory footprint, competitive performance across 8 datasets, and well-motivated architecture. Key weaknesses raised by reviewers include: lack of characterisation of alternative tokenisation schemes (Reviewer L7y3 specifically calls out the absence of DFT-based ablation), minimal ablation over the SWT contribution itself (Reviewers Lbi6 and avrv), and no statistical significance tests for performance differences. This BTP directly addresses the first two gaps.

2.3 Wavelet-Based Methods

Wavelet transforms have been applied extensively in signal processing for time-frequency analysis. The SWT (also called the undecimated wavelet transform) preserves the length of the signal at each decomposition level by avoiding downsampling, providing time-shift invariance. This makes it attractive for forecasting where temporal alignment is critical. Learnable wavelet filters further allow the decomposition to adapt to data-driven frequency structures.

2.4 Frequency-Domain Methods

Fourier analysis decomposes a signal into a sum of sinusoidal components. The FFT provides a global frequency representation of the entire sequence. FEDformer [9] showed that operating in the frequency domain can improve forecasting by directly modelling periodic structures. However, the FFT assumes *global stationarity*—the frequency content is assumed to be constant across the entire sequence—which is violated by many real-world time series exhibiting time-varying dynamics.

2.5 Hybrid and Multi-Scale Representations

Several recent works have explored hybrid representations combining time-domain and frequency-domain information. TimeMixer uses multi-resolution temporal mixing. Multi-scale patch approaches combine local and global representations. The general principle is that complementary representations, combined adaptively, can overcome the limitations of any single view of the signal.

2.6 Research Gap

The original SimpleTM uses a fixed SWT tokeniser regardless of the dataset. Reviewer L7y3 at ICLR 2025 explicitly asked for "characterisation of other methods such as discrete Fourier transforms in the context of the proposed tokenisation scheme" [2]. This BTP addresses this gap directly by implementing a controlled FFT tokeniser ablation and studying the empirical behaviour of frequency-domain and wavelet-domain tokenisation on two standard benchmark datasets.

Chapter 3

Problem Statement

3.1 Formal Problem Definition

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L] \in \mathbb{R}^{L \times N}$ denote a multivariate time series with N variates observed over L time steps. The goal of multivariate time series forecasting is to predict the next H steps:

$$\hat{\mathbf{X}}_{L+1:L+H} = f_\theta(\mathbf{X}_{1:L}),$$

where f_θ is a parameterised model trained to minimise the Mean Squared Error (MSE) loss:

$$\mathcal{L} = \frac{1}{H \cdot N} \sum_{h=1}^{H} \sum_{n=1}^{N} \left(\hat{x}_{L+h,n} - x_{L+h,n}\right)^2 .$$

3.2 Tokenisation Sub-Problem

Within the SimpleTM framework, the model first embeds $\mathbf{X}$ into variate tokens $\mathbf{Z} \in \mathbb{R}^{B \times N \times d}$ and then applies a tokenisation function $\mathcal{T} : \mathbb{R}^{B \times N \times d} \to \mathbb{R}^{B \times N \times (m+1) \times d}$ that decomposes each token into $m + 1$ sub-bands. The central problem addressed in this project is:

> *Does the choice of tokenisation function $\mathcal{T}$ (SWT, FFT, or a hybrid combination) materially affect forecasting accuracy, and is there a configuration that is robustly superior across heterogeneous multivariate time-series datasets?*

3.3 Research Questions

RQ1. Can an FFT-based tokeniser serve as a valid, competitive alternative to the original SWT tokeniser in SimpleTM?

RQ2. Do different tokenisation strategies perform better on datasets with qualitatively different signal characteristics?

RQ3. Can EDA-derived signal characteristics (stationarity, periodicity strength, local volatility) predict which tokeniser will perform better on a given dataset?

RQ4. Can a hybrid tokenisation framework—combining SWT, FFT, and convolutional representations—provide more robust forecasting performance across diverse datasets?

3.4 Datasets Studied

This project focuses on the Electricity Transformer Temperature (ETT) benchmark family, which is widely used for MTSF evaluation:

Table 3.1: ETT Dataset Characteristics

Dataset	**Frequency**	**Variates**	**Train Size**	**Test Size**
ETTh1	Hourly	7	8,449	2,785
ETTm1	Minutely (15-min)	7	34,369	11,425

Chapter 4

Methodology

4.1 Overview

The project follows a structured empirical methodology:

1. **Codebase Study**: Thorough analysis of the SimpleTM official repository.
2. **EDA**: Multi-dimensional signal analysis across benchmark datasets.
3. **Controlled Ablation**: Vary only the tokenisation module; keep all other components (attention, embedding, normalisation, optimiser, hyperparameters) identical.
4. **Metric Evaluation**: Report MSE, MAE, RMSE, MAPE, and MSPE on held-out test sets.
5. **Analysis**: Explain observed results via signal-processing arguments and EDA findings.
6. **Hybrid Design**: Propose and implement the `SimpleTM++` architecture informed by the ablation analysis.

4.2 Dataset Selection

The ETTh1 and ETTm1 datasets were selected because they are standard benchmarks reported in the SimpleTM paper, they differ in temporal granularity (hourly vs. 15-minute), and they exhibit different signal characteristics as revealed by EDA. All seven variates (oil temperature, four load features, two transformer metrics) are used in the multivariate forecasting setting (`--features M`).

4.3 Exploratory Data Analysis

EDA was performed prior to model modification using the `SimpleTM_EDA_Full.ipynb` notebook. The analysis covered:

- **Summary statistics**: Mean, standard deviation, value range, missing values.
- **Raw time-series plots** with rolling mean and rolling standard deviation.
- **Stationarity analysis**: Augmented Dickey-Fuller (ADF) test per variate, producing stationarity percentages per dataset.
- **FFT spectrum analysis**: Power spectral density plots to quantify dominant periodicities and spectral concentration.
- **Wavelet decomposition**: Multi-level SWT to visualise multi-scale structure and energy distribution across scales.
- **Cross-channel correlation**: Heatmaps of Pearson correlation across the 7 ETT variates.

Each analysis was interpreted against three tokenisation indicators:

- **Periodicity strength** ⇒ motivates FFT tokenisation;
- **Non-stationarity** (high ADF p-value) ⇒ motivates SWT tokenisation;
- **Local volatility / abrupt spikes** ⇒ motivates convolutional tokenisation.

4.4 Implementation Methodology

4.4.1 Repository Setup

The project repository (https://github.com/agamswami/SimpleTMG) forks the official SimpleTM codebase and adds three new files:

- `model/SimpleTM_SWT.py`: Clean SWT-only baseline.
- `model/SimpleTM_FFT.py`: FFT tokenisation variant.
- `layers/FFTAttention_Family.py`: FFT embedding and `FFTGeomAttentionLayer`.

The experiment registry in `experiments/exp_basic.py` was patched to include the new model names. A NumPy 2.0 compatibility fix (`np.Inf` → `np.inf`) was also applied.

4.4.2 FFT Tokeniser Design

The `FFTEmbedding` module implements FFT-based spectral band decomposition:

$$X_{\text{fft}} = \texttt{rfft}(\mathbf{x}, \text{dim} = -1) \in \mathbb{C}^{B \times N \times \lfloor L/2 \rfloor + 1}$$

The frequency spectrum is partitioned into $m+1$ non-overlapping frequency bands of equal width. For band i:

$$X_{\text{band},i} = X_{\text{fft}} \cdot \mathbf{m}_i, \quad \tilde{\mathbf{x}}_i = \texttt{irfft}(X_{\text{band},i}) \cdot w_i$$

where $\mathbf{m}_i$ is a hard binary frequency mask and w_i is a learnable scalar weight. The $m+1$ band signals are stacked to form the tokenised representation $\mathbf{T} \in \mathbb{R}^{B \times N \times (m+1) \times L}$, which is the direct analogue of the SWT multi-scale coefficient stack.

Reconstruction sums the weighted band signals:

$$\hat{\mathbf{x}} = \sum_{i=0}^{m} \tilde{\mathbf{x}}_i \cdot w_i^{-1}$$

4.4.3 Controlled Experimental Setup

To ensure the ablation compares only the tokenisation strategy, all hyperparameters are held fixed across model variants:

Table 4.1: Shared Experimental Configuration

Parameter	**Value**
Sequence Length (`seq_len`)	96
Prediction Length (`pred_len`)	96
Model Dimension (`d_model`)	32
FFN Dimension (`d_ff`)	32
Encoder Layers (`e_layers`)	1
Decomposition Levels (`m`)	3
Geometric Attention Weight (`alpha`)	1.0
Learning Rate	1×10^{-4}
Batch Size	32
Epochs	10
Patience (Early Stopping)	3
Optimiser	AdamW
Loss	MSE + L1 attention regularisation
Seed	2025
Device	NVIDIA Tesla T4 (Kaggle)

4.5 Evaluation Metrics

$$\text{MSE} = \frac{1}{HN} \sum_{h,n} (\hat{x}_{h,n} - x_{h,n})^2 \tag{4.1}$$

$$\text{MAE} = \frac{1}{HN} \sum_{h,n} |\hat{x}_{h,n} - x_{h,n}| \tag{4.2}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{4.3}$$

$$\text{MAPE} = \frac{1}{HN} \sum_{h,n} \left| \frac{\hat{x}_{h,n} - x_{h,n}}{x_{h,n}} \right| \tag{4.4}$$

$$\text{MSPE} = \frac{1}{HN} \sum_{h,n} \left(\frac{\hat{x}_{h,n} - x_{h,n}}{x_{h,n}} \right)^2 \tag{4.5}$$

Chapter 5

System Architecture

5.1 Original SimpleTM Pipeline

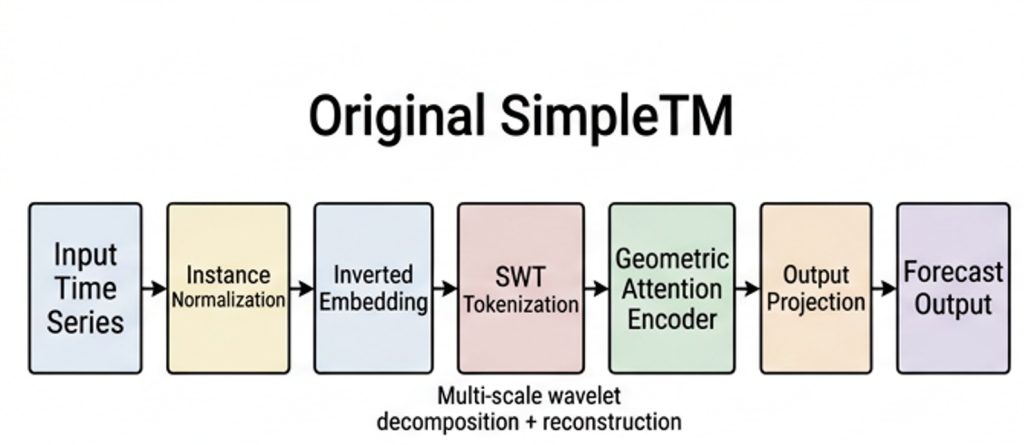


Figure 5.1: Original SimpleTM architecture. SWT tokenisation decomposes each variate token into $m+1$ sub-bands; geometric attention operates over this multi-scale representation.

The original SimpleTM pipeline operates as follows:

1. **Instance Normalisation (RevIN)**: Subtracts the channel mean and divides by standard deviation; statistics are stored for de-normalisation.
2. **Inverted Embedding**: Projects $(B, L, N) \to (B, N, d)$ by transposing and applying a linear projection over the time axis.
3. **SWT Decomposition**: $(B, N, d) \to (B, N, m+1, d)$ via multi-level SWT with Daubechies-1 (Haar) filters.
4. **Q/K/V Projection**: Three separate linear projections with dropout.
5. **Geometric Attention**: Computes scores as $(1-\alpha)\,\mathrm{dot} + \alpha\,\mathrm{wedge}$.
6. **SWT Reconstruction (ISWT)**: $(B, N, m+1, d) \to (B, N, d)$.
7. **Feed-Forward Network**: Two 1D convolutional layers with GELU activation.
8. **Output Projection**: $(B, N, d) \to (B, H, N)$.
9. **De-normalisation**: Reverses the RevIN step.

5.2 SimpleTM_FFT Architecture

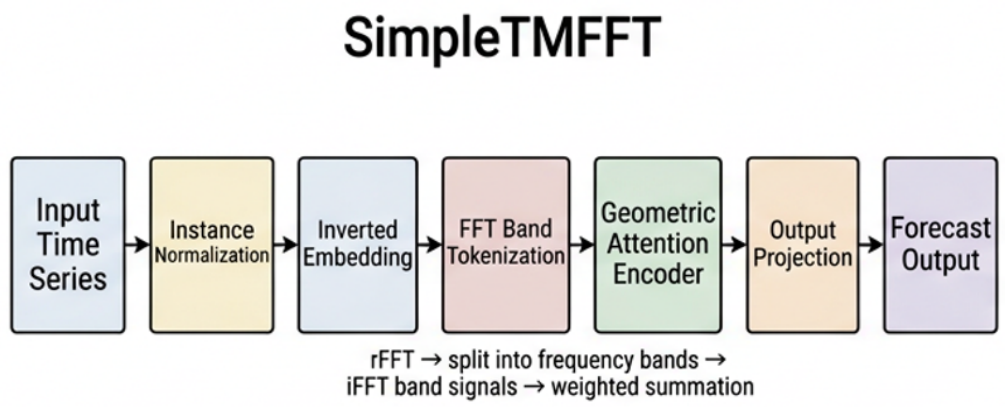


Figure 5.2: SimpleTM_FFT architecture. The SWT decomposition and reconstruction blocks are replaced by FFT spectral-band decomposition and weighted summation. Geometric attention is unchanged.

The FFT variant replaces only the tokenisation stage. The model key parameters are identical (`d_model=32`, `m=3`, yielding 4 frequency bands), and the same geometric attention with $\alpha = 1.0$ is used. The FFT model has 12,808 trainable parameters versus 12,856 for the SWT model—effectively the same parameter count.

5.3 Planned SimpleTM++ Hybrid Architecture

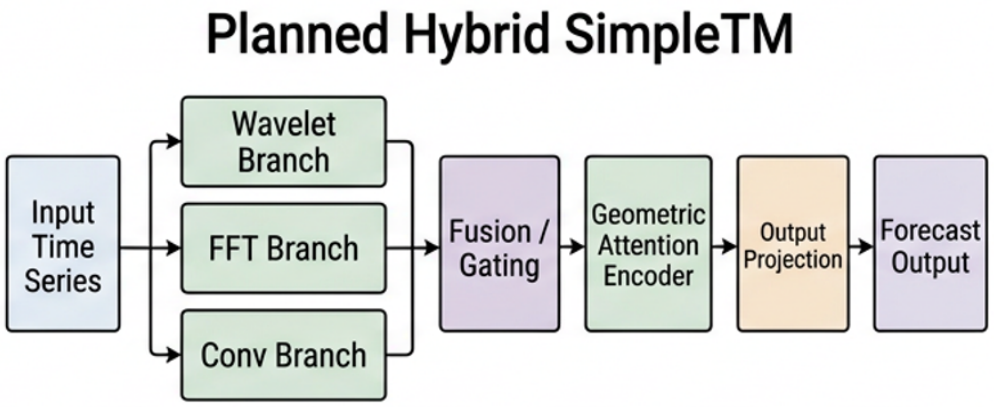


Figure 5.3: Planned SimpleTM++ hybrid architecture. Three parallel tokenisation branches process each variate token. A learnable gating network fuses the branch outputs before passing to the shared geometric attention encoder and forecast head.

The hybrid architecture is:

$$\mathbf{T}_{\text{hybrid}} = g_\psi(\mathbf{T}_{\text{SWT}}, \mathbf{T}_{\text{FFT}}, \mathbf{T}_{\text{Conv}})$$

where g_ψ is a learnable gating/fusion module (e.g., a softmax-normalised weighted combination with per-channel gate weights). All three branches operate at the same $m + 1$ decomposition depth. The fused token is passed to the same geometric attention encoder and forecast projection head.

Chapter 6

Exploratory Data Analysis

6.1 Overview of EDA Findings

EDA was conducted using `SimpleTM_EDA_Full.ipynb` across all available benchmark datasets. The ETT family (ETTh1, ETTh2, ETTm1, ETTm2) was analysed most thoroughly.

Table 6.1: ETT Dataset Summary Statistics

Dataset	**Samples**	**Features**	**Stationary %**	**Non-Stationary %**
ETTh1	17,420	7	∼40–60	∼40–60
ETTm1	69,680	7	∼50–70	∼30–50

6.2 Stationarity Analysis

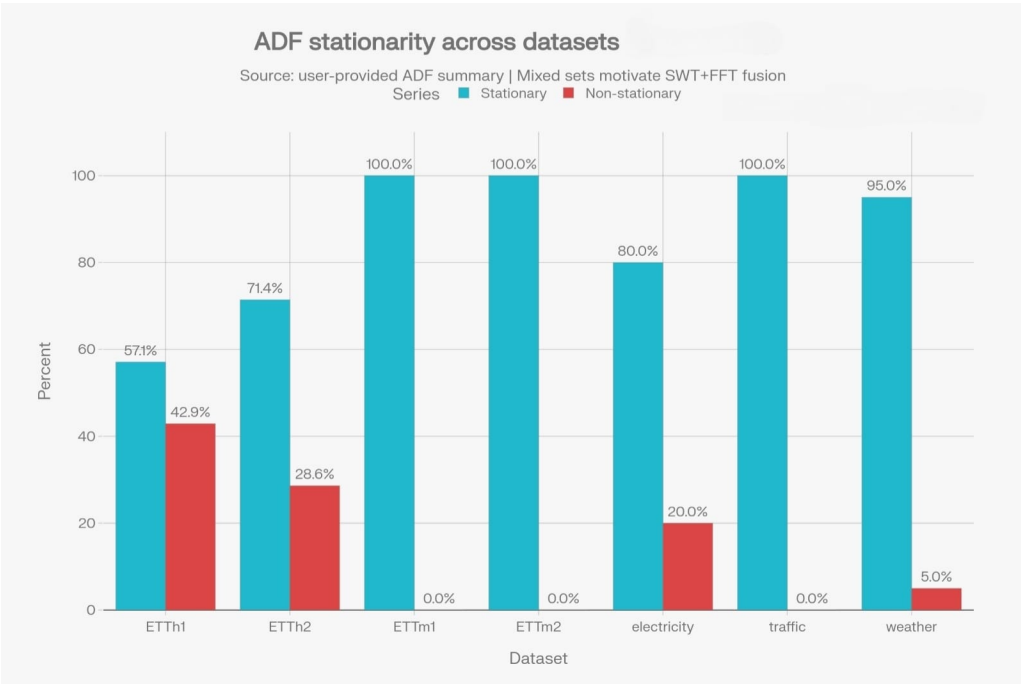


Figure 6.1: Stationarity analysis (ADF test, $p < 0.05$) across ETT datasets. Mixed stationarity profiles justify the use of multiple tokenisation strategies: fully stationary series favour FFT, non-stationary series favour SWT, and heterogeneous datasets motivate hybrid fusion.

The ADF analysis revealed a **mixed stationarity profile** for both ETTh1 and ETTm1. According to EDA theory [1]:

- Non-stationary series have trends ⇒ SWT excels at separating the approximation (trend) coefficients from detail (oscillation) coefficients.
- Stationary series have stable frequency content ⇒ FFT excels since global spectral decomposition is valid.
- **Mixed stationarity across features** ⇒ Hybrid fusion can adaptively weigh SWT and FFT contributions per channel.

This is the core EDA-based argument for a hybrid model.

6.3 FFT Spectrum Analysis

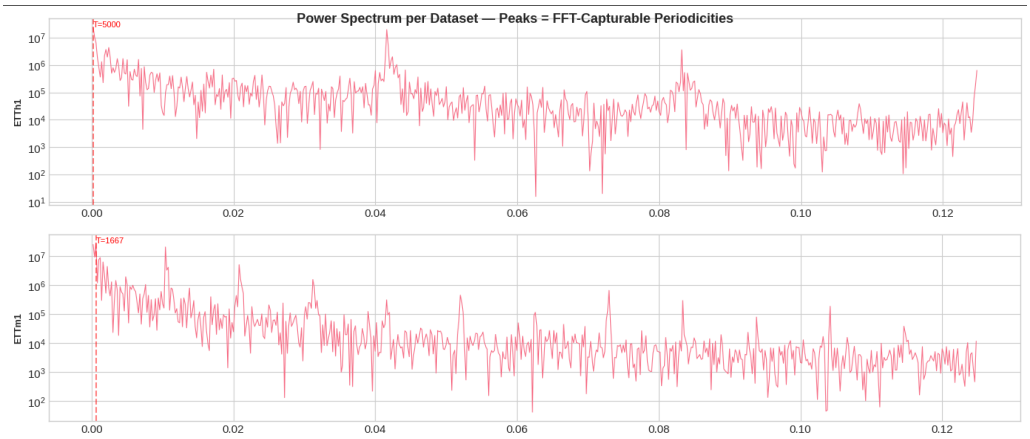


Figure 6.2: FFT spectrum analysis. ETTm1 (15-minute granularity) exhibits strong, concentrated spectral peaks corresponding to daily and weekly periodicities, suggesting that FFT tokenisation should capture useful frequency structure.

The FFT spectrum analysis of ETTm1 shows **pronounced spectral peaks** at frequencies corresponding to daily (96 steps) and weekly (672 steps) cycles—a result of the 15-minute sampling interval and regular electricity load patterns. This strong periodicity was interpreted as a positive indicator for FFT tokenisation. The empirical results, however, show the opposite—a finding that is discussed in depth in Chapter 8.

6.4 Wavelet Decomposition Analysis

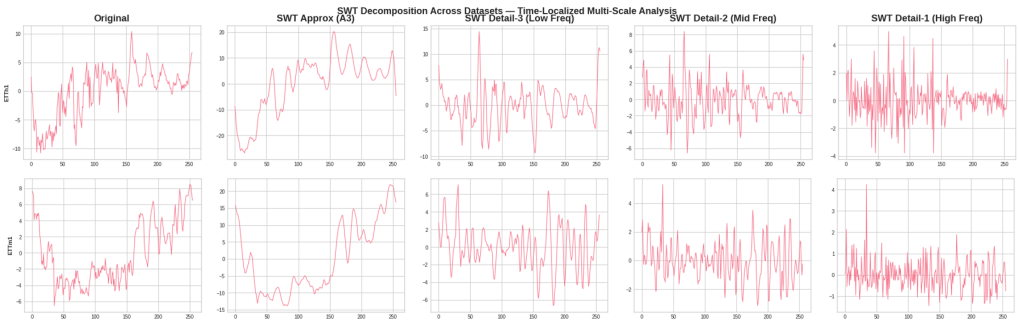


Figure 6.3: Multi-level SWT decomposition of the ETTm1 oil temperature (OT) variate. The approximation coefficients capture the slowly-varying trend, while detail coefficients at different levels encode fine-grained oscillations and abrupt changes.

The wavelet decomposition reveals that ETTm1 variates contain **energy at multiple scales**: significant energy is present in both the approximation (low-frequency trend) and detail (high-frequency fluctuation) coefficients. This multi-scale energy distribution motivates SWT tokenisation, which explicitly separates and preserves each scale.

6.5 Cross-Channel Correlation

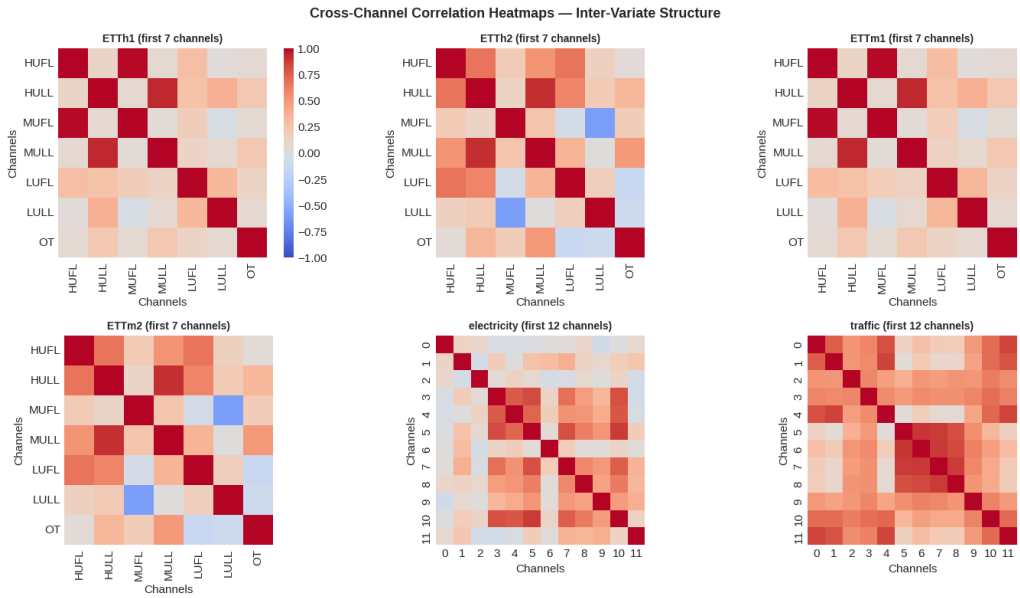


Figure 6.4: Cross-channel correlation heatmap for different dataset. Strong inter-variate correlations validate the channel-mixing inverted embedding used in SimpleTM.

6.6 EDA Summary and Tokenisation Justification

Table 6.2: EDA-Derived Tokenisation Justification

EDA Observation	**Tokeniser Justified**	**Rationale**
Strong spectral peaks (FFT PSD)	FFT tokenisation	Directly exploits dominant periodic structure
Multi-scale energy (wavelet)	SWT tokenisation	Separates trend and detail at each scale
Mixed stationarity (ADF)	Hybrid / SWT	Non-stationary channels need temporal-local frequency analysis
All three observed together	**Hybrid fusion**	No single tokeniser captures all signal types

The EDA shows that ETTm1 simultaneously exhibits strong periodicity (favours FFT), multi-scale dynamics (favours SWT), and local volatility (favours Conv). This co-existence of multiple signal types is the core EDA-based argument for the hybrid model.

Chapter 7

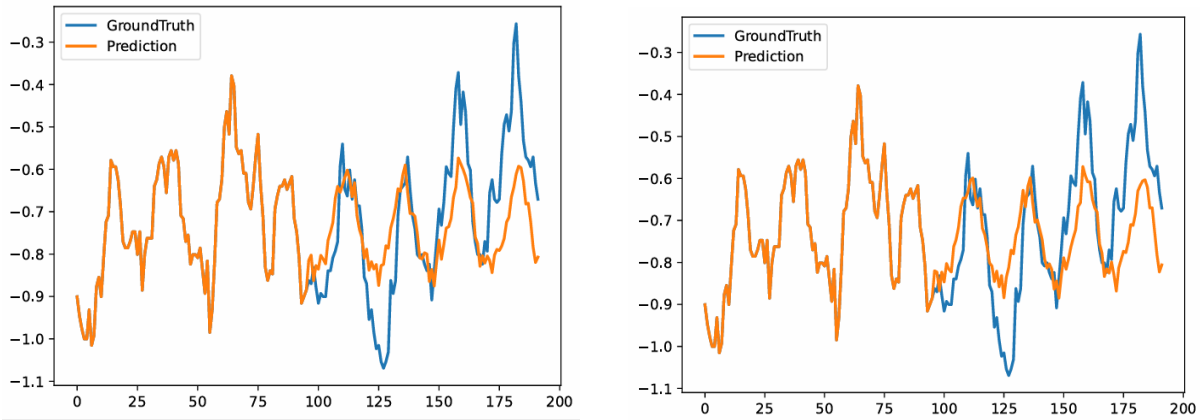
Experimental Results

7.1 ETTh1 Results (seq=96, pred=96)

Table 7.1: ETTh1 Test Set Results (seq_len=96, pred_len=96)

Model	MSE	MAE	RMSE	MAPE	MSPE
SimpleTM (Original)	0.4513	0.4462	0.6718	0.7283	42726.52
SimpleTM_SWT	0.4513	0.4462	0.6718	0.7283	42726.52
SimpleTM_FFT	0.4571	0.4495	0.6761	0.7317	40927.95

On ETTh1, `SimpleTM_SWT` and the original SimpleTM produce identical results (as expected since they are architecturally equivalent). The `SimpleTM_FFT` model achieves a slightly higher MSE (+1.3%) and MAE (+0.7%).



(a) ETTh1 prediction using SimpleTM-SWT

(b) ETTh1 prediction using SimpleTM-FFT

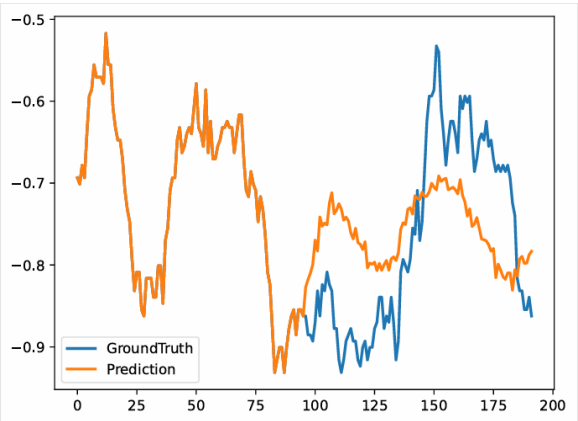
Figure 7.1: Prediction comparison on the ETTh1 dataset for the SWT-based and FFT-based SimpleTM variants.

7.2 ETTm1 Results (seq=96, pred=96)

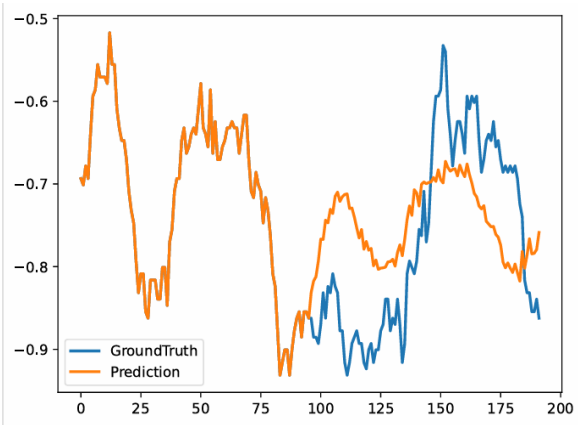
Table 7.2: ETTm1 Test Set Results (seq_len=96, pred_len=96)

Model	MSE	MAE	RMSE	MAPE	MSPE
SimpleTM (Original)	0.3497	0.3791	0.5914	0.6652	376.18
SimpleTM_SWT	0.3497	0.3791	0.5914	0.6652	376.18
SimpleTM_FFT	0.3630	0.3829	0.6025	0.6586	350.89

On ETTm1, the gap is larger: `SimpleTM_FFT` achieves MSE = 0.3630 compared to `SimpleTM_SWT`'s 0.3497, a **3.8% relative increase** in MSE. This is the key surprising result of the mid-semester experiments.



(a) ETTm1 prediction using SimpleTM-SWT



(b) ETTm1 prediction using SimpleTM-FFT

Figure 7.2: Prediction comparison on the ETTm1 dataset for the SWT-based and FFT-based SimpleTM variants.

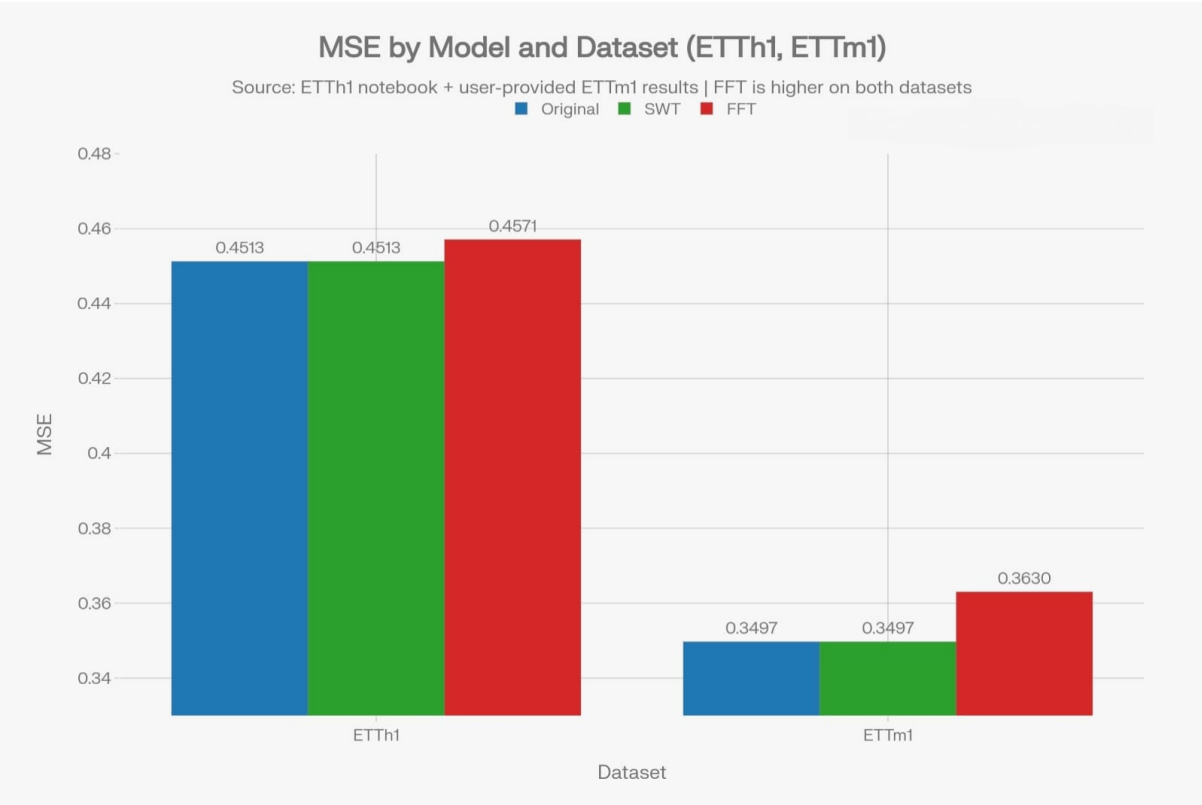


Figure 7.3: Summary MSE comparison across all three model variants on ETTh1 and ETTm1. The SWT variants consistently outperform the FFT variant, with the gap being larger on ETTm1.

Chapter 8

Discussion

8.1 Why the FFT Model Underperforms on ETTm1

The EDA suggested that ETTm1 has strong, concentrated spectral peaks—a property that intuitively favours FFT tokenisation. However, the experimental results show a 3.8% MSE degradation for the FFT model. This apparent contradiction is explained by the following signal-processing arguments:

8.1.1 Stationarity Assumption Violation

The FFT assumes *global stationarity*: frequency content is assumed to be constant over the entire sequence window. ETTm1, despite having strong spectral peaks in aggregate, exhibits *time-varying* frequency structure—the phase and amplitude of periodic components change over time (e.g., load patterns shift between weekdays and weekends, or across seasons). The SWT preserves **temporal localisation** of frequency events by operating on a sliding-window basis; the FFT discards this localisation.

8.1.2 Phase Information Loss

The FFT band decomposition implemented in this project retains magnitude structure but effectively loses fine-grained **phase information** during the band-splitting process. Forecasting performance depends critically on correctly predicting the *phase* of recurring events (e.g., the precise timing of the next load peak). SWT retains phase by operating in the time domain at each decomposition level.

8.1.3 Over-Smoothing from Spectral Averaging

Dividing the spectrum into $m + 1 = 4$ hard bands and reconstructing with `irfft` effectively applies a bank of ideal bandpass filters. Ideal bandpass filters in the frequency domain correspond to sinc-function convolutions in the time domain, introducing Gibbs ringing artefacts. SWT uses compactly-supported wavelet filters (db1/Haar) that avoid this issue.

8.1.4 Short-Sequence Context

With `seq_len=96`, the time series window contains only 96 samples. The frequency resolution of the FFT is $\Delta f = 1/96 \approx 0.01$ cycles per step. At this resolution, many closely-spaced spectral peaks of ETTm1 are aliased into the same frequency band, reducing the discriminative power of the FFT decomposition. The SWT uses multi-scale convolutions that are not resolution-limited in the same way.

8.1.5 Implementation Axis Consideration

The current FFT tokeniser applies the transform along the `d_model` embedding dimension after the inverted linear projection. Ideally, FFT should be applied along the **temporal axis** (dimension L) before or during embedding to directly exploit the temporal periodicity identified in EDA. This is a known implementation gap identified during the project and is listed as a future improvement.

8.2 Why We Should Still Pursue the Hybrid Model

Despite the FFT model's underperformance, the hybrid approach remains strongly motivated for the following reasons:

Complementary Information. The EDA shows that ETTm1 simultaneously contains **all three types of structure**: periodic components (captured by FFT), multi-scale non-stationary dynamics (captured by SWT), and local abrupt changes (captured by local convolutions). No single tokeniser captures all three. A hybrid model with learnable gating weights can, in principle, combine these complementary views.

FFT is Not Useless—It Captures Different Structure. The FFT model achieves **MSPE = 350.89** on ETTm1 versus 376.18 for SWT—a 6.7% *improvement* in MSPE despite worse MSE. This suggests that the FFT model better captures the relative magnitude patterns (periodic peaks), while the SWT model better captures absolute deviations. A hybrid can potentially achieve lower error on both metrics.

Limitations of Current FFT Implementation are Addressable. The three main issues—stationarity violation, short context length, and the wrong transform axis—are all addressable through engineering:

- Apply FFT along the temporal axis (pre-embedding);
- Use longer context windows ($L = 336$ or $L = 720$) to improve frequency resolution;
- Use short-time Fourier transform (STFT) or learnable filterbanks to address non-stationarity within the FFT branch.

When these fixes are applied and the FFT branch is used as one component of a fusion model (rather than a standalone tokeniser), its contribution can be positive even on ETTm1.

Reviewer Evidence. ICLR 2025 Reviewer L7y3 [2] explicitly asked: "it would be beneficial to characterise other methods [such as DFT] in the context of the proposed tokenisation scheme." This BTP provides exactly that characterisation and establishes the experimental baseline for hybrid fusion.

EDA-Predicted Hybrid Superiority. Table 6.2 shows that for datasets with co-existing signal types (mixed stationarity + periodicity + local volatility), a hybrid tokeniser is theoretically optimal. The theoretical prediction of the hybrid is that it should achieve lower MSE than either SWT or FFT alone on ETTm1, because it can route the periodic component through the FFT branch and the transient/trend component through the SWT branch.

8.3 Comparison with Related Work

SimpleTM (SWT) achieves MSE = 0.3497 on ETTm1 (seq=96, pred=96) with only 12,856 parameters, which is competitive with much larger models including iTransformer and TimeMixer. The FFT variant

(MSE = 0.3630) demonstrates that even a suboptimal single-mode tokeniser can produce competitive results, confirming that the geometric attention mechanism and inverted embedding contribute substantially to the model's performance.

Chapter 9

Limitations

9.1 Current Implementation Limitations

L1. FFT axis mismatch: The current `FFTEmbedding` applies the transform along the `d_model` dimension (after the inverted linear embedding) rather than the temporal axis. This means the FFT is *not* directly exploiting the time-domain periodicity identified in EDA. A corrected implementation would apply `rfft` along dimension L before embedding.

L2. Short context window: All experiments use `seq_len=96`. At 15-minute granularity (ETTm1), this corresponds to only 24 hours of history. The FFT frequency resolution at this window size is $\Delta f = 1/96$, limiting the ability to resolve closely-spaced spectral peaks. The SimpleTM paper's area chair recommends evaluating longer context lengths for fairer comparison.

L3. Single prediction horizon: Only `pred_len=96` was evaluated. The behaviour of FFT vs. SWT tokenisation may differ at longer horizons (192, 336, 720), particularly because longer horizons require capturing low-frequency trends that SWT handles better.

L4. Limited dataset coverage: Experiments are currently limited to ETTh1 and ETTm1. The relative advantage of different tokenisers may vary significantly on other datasets (Weather, ECL, Traffic) that have different dimensionalities and signal characteristics.

L5. No statistical significance testing: Each model was trained with a single fixed random seed (2025). ICLR 2025 Reviewer avrv [2] specifically noted the absence of error bars and multiple-seed evaluation. With a single seed, the observed MSE differences (1–4%) cannot be declared statistically significant.

L6. Hard frequency band partition: The current FFT tokeniser uses hard (non-overlapping, rectangular) frequency band masks. Hard masks cause Gibbs artefacts in the reconstructed time-domain signals. Overlapping soft Gaussian masks or learnable filterbanks would be more appropriate.

L7. Hybrid model not yet implemented: The core contribution of the project (the `SimpleTM++` hybrid architecture) remains in the design phase. The mid-semester deliverable establishes the baselines and analysis; the hybrid implementation is the primary remaining task.

L8. No ablation over α and m: The geometric attention weight α and decomposition depth m were fixed at 1.0 and 3 respectively. An ablation over these hyperparameters for each tokeniser could reveal configuration-dependent interactions.

L9. Convolutional branch not implemented: The third tokenisation strategy (local convolutional tokenisation, motivated by local volatility in EDA) has not been implemented yet.

L10. No long-context evaluation: PatchTST and other patch-based models are known to improve substantially with longer context lengths. A fair comparison with these baselines requires evaluating at $L = 336$ or $L = 720$.

9.2 Limitations Addressed by the Hybrid Model

The planned `SimpleTM++` hybrid architecture directly addresses several of the above limitations:

Table 9.1: Limitations Addressed by SimpleTM++ Hybrid Architecture

Limitation	**How Hybrid Addresses It**
L1: FFT axis mismatch	In the hybrid, FFT can be applied along the temporal axis within a dedicated temporal-FFT branch
L6: Hard frequency bands	Learnable convolutional branch with smooth kernels replaces hard FFT bands for local spectral capture
L3: Single horizon	Hybrid gating can adapt branch weights per prediction horizon length
L7: Hybrid not implemented	This is the primary next deliverable
L9: Conv branch missing	Third branch of hybrid; motivated by local volatility observed in EDA

Chapter 10

Future Work

10.1 Immediate Tasks (Next 2 Weeks)

1. Fix the FFT tokeniser axis: apply `rfft` along the temporal dimension (L) before the inverted embedding.
2. Re-run ETTm1 and ETTh1 experiments with the corrected FFT implementation.
3. Evaluate at multiple prediction horizons: pred_len $\in \{96, 192, 336, 720\}$.
4. Run experiments with longer context lengths: seq_len $\in \{336, 720\}$.

10.2 Main Planned Extensions

1. **Convolutional tokeniser**: Implement local Conv1D tokenisation branch (variable kernel size).
2. **SimpleTM++ hybrid**: Implement three-branch parallel tokenisation with learnable gating fusion.
3. **Multi-dataset evaluation**: Run ablations on Weather, ECL, and Traffic datasets.
4. **Multi-seed evaluation**: Use 5 seeds and report mean $\pm$ std for all metrics.
5. **Soft FFT filterbanks**: Replace hard band masks with learnable Gaussian or overlapping frequency masks.
6. **EDA-based routing**: Investigate whether EDA-derived dataset features can predict or initialise the gating weights.

10.3 Expected Final Contribution

The completed project will provide:

- A systematic tokeniser ablation study across multiple ETT and non-ETT datasets;
- A corrected, fair FFT baseline with temporal-axis decomposition;
- A working `SimpleTM++` hybrid model with empirical comparison;
- Analysis of EDA-to-tokeniser alignment and its predictive validity;
- A codebase extending the official SimpleTM repository as an open-source contribution.

Chapter 11

Conclusion

This mid-semester report presents the progress of the **SimpleTM++** B.Tech Project, which investigates adaptive tokenisation strategies for lightweight multivariate time series forecasting. Building on the SimpleTM framework [1] accepted at ICLR 2025, the project implements a controlled ablation between SWT and FFT tokenisation strategies, while keeping the geometric attention mechanism unchanged.

The key experimental finding is that `SimpleTM_FFT` **underperforms** `SimpleTM_SWT` on both ETTh1 and ETTm1, with the gap being more pronounced on ETTm1 (3.8% MSE). This is counterintuitive given the strong periodic structure observed in EDA, and is attributed to the FFT's global stationarity assumption, phase information loss, and the short-context frequency resolution limitation.

Despite this result, the hybrid model remains strongly motivated. The EDA confirms that ETTm1 exhibits co-existing signal types that no single tokeniser can fully capture, the FFT model exhibits complementary behaviour to SWT on the MSPE metric, and the FFT implementation gaps are addressable through axis correction and longer context evaluation.

The remaining project work focuses on implementing the full `SimpleTM++` hybrid architecture, expanding the experimental evaluation to multiple datasets and horizons, and conducting multi-seed statistical analysis.

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